LOGICAL CONSTRAINTS ON THE EVENTUAL BEHAVIOR OF SOME CELLULAR AUTOMATA

Milan Rosko

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ORCID: 0009-0003-1363-7158

Abstract

We develop in Rocq a finite-witness method for refuting eventual periodicity in one-dimensional cellular automata, applied to the canonical single-seed trajectory of Rule 30. Instead of global asymptotic arguments, we reduce any periodicity hypothesis to a bounded Boolean constraint system inside a backward light cone. This system consists of local rule updates, overlap relations between neighboring triples, and a period-closing condition. We formalize the necessary local infrastructure (truth-table inversions, finite-speed dependence, and shifted-cone propagation) and use it to eliminate all candidate witnesses. The resulting obstruction principle is finite and mechanically checkable.

1 INTRODUCTION

1.1 CELLULAR SEMANTICS

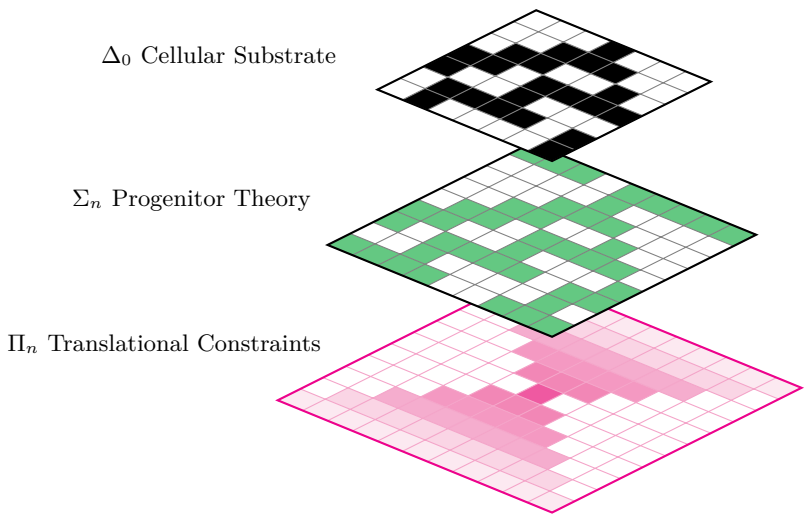

Figure 1. The recursion-theoretic argument in miniature. Top: Δ_0 cellular substrate. Middle: Σ_n preimage analysis. Bottom: Π_n periodicity hypothesis inheriting the obstruction.

Remark 1.1.1. Rule 30 ($\mathcal{R}_{30}$) is one of the simplest one-dimensional binary cellular automata with visibly complex behavior [Wolfram, 1983, 2002]. Starting from a single active cell, it generates a cellular "spacetime" diagram whose left side exhibits regular structure while the central and right regions appear irregular. This contrast has made it a standard example in discussions of emergence, pseudorandomness, and the limits of prediction in deterministic systems [Wolfram, 2002]. Yet even in this classical setting,

basic structural questions remain delicate when one insists on exact proofs of the *first order*, cf. Boolos et al. [2012], rather than statistical or visual evidence.

In this paper we study a sharply localized version of that problem. Rather than asking for global periodicity of full rows or of wide finite windows, we focus on the center column of the canonical single-seed trajectory. Writing

$$u_{t,x} \tag{1}$$

for the Rule 30 spacetime and

$$c(t) := u_{t,0} \tag{2}$$

for the center value at time t, the question is whether the binary sequence $c(t)$ can become eventually periodic. The eventual periodicity of the center column is one of the open questions highlighted and revisited extensively in Wolfram [2002, Chapter 6]. This center-only formulation isolates the most rigidly defined observable in the standard single-seed evolution while avoiding the additional combinatorial overhead of global strip dynamics.

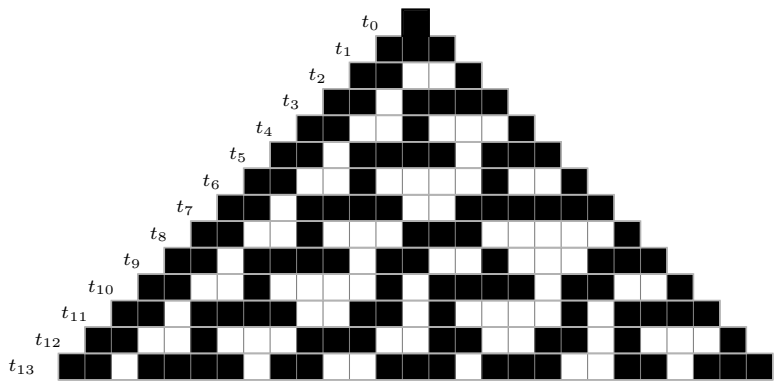

Figure 2. The canonical single-seed $\mathcal{R}_{30}$ spacetime. Row $t = 0$ contains the single active cell; each subsequent row is produced by the update rule $\mathcal{R}_{30}(a, b, c) = a \oplus (b \vee c)$. The light cone expands by one cell per side per step.

Definition 1.1.2 (Rule 30 trajectory and center column [Wolfram, 1983])**.** Let

$$\mathcal{R}_{30}(a, b, c) := a \oplus (b \vee c), \quad a, b, c \in \{0, 1\}. \tag{3}$$

A Rule 30 *spacetime* is a family of bits

$$u_{t,x} \in \{0, 1\} \quad (t \in \mathbb{N},\ x \in \mathbb{Z}) \tag{4}$$

satisfying the local *update rule*, the 8 evolution laws for cells: if □ represents 0 and ■ represents 1, as in Figure 2, then at time $t + 1$:

$$\begin{array}{cccccccc} \blacksquare\blacksquare\blacksquare & \blacksquare\blacksquare\square & \blacksquare\square\blacksquare & \blacksquare\square\square & \square\blacksquare\blacksquare & \square\blacksquare\square & \square\square\blacksquare & \square\square\square \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ \square & \square & \square & \blacksquare & \blacksquare & \blacksquare & \blacksquare & \square \end{array} \tag{5}$$

formally,

$$u_{t+1,x} = \mathcal{R}_{30}(u_{t,x-1}, u_{t,x}, u_{t,x+1}) \quad \text{for all } t \in \mathbb{N},\ x \in \mathbb{Z}. \tag{6}$$

In this paper we work with the *canonical single-seed trajectory*, defined by the initial row

$$u_{0,x} = \begin{cases} 1, & x = 0, \\ 0, & x \neq 0. \end{cases} \tag{7}$$

The *center column* of this trajectory is the *binary sequence*

$$c : \mathbb{N} \to \{0, 1\}, \quad c(t) := u_{t,0}. \tag{8}$$

1.2 QUESTION AND ANSWER

Definition 1.2.1 (Center periodicity)**.** Let $u_{t,x}$ and $c(t) := u_{t,0}$ be as above. For integers $T \geq 0$ and $P > 0$, we say that the center column is *periodic from time T with period P* if

$$c(t + P) = c(t) \quad \text{for all } t \geq T. \tag{9}$$

It is *eventually periodic* if such integers T and P exist.

Definition 1.2.2 (Question)**.** The problem addressed in the paper is narrow and of central interest to the science of dynamical systems. We examine the most rigidly specified observable in the canonical single-seed evolution: the center column

$$c(t) := u_{t,0} \tag{10}$$

and ask whether this binary sequence can become *eventually periodic*. That is, do there exist integers $T \geq 0$ and $P > 0$ such that

$$c(t + P) = c(t) \quad \text{for all } t \geq T? \tag{11}$$

Theorem 1.2.3 (Answer)**.** Our *Ansatz* is *proof by induction*, with the inductive step by *reductio ad absurdum*, checked in an explicit Calculus of Constructions.

The core argument is *constructive* in the sense of the Brouwer–Heyting–Kolmogorov Interpretation [Troelstra and van Dalen, 1988]: every refutation step produces an explicit finite witness or contradiction, and no step requires deciding an unbounded search. Since eventual periodicity is a classical question—it asks whether certain integers T and P exist—the final statement is naturally discharged under the *Law of the Excluded Middle* (lem). In the mechanization, this appears as the sole use of `Classical_Prop.classic` in the tail anti-periodicity layer, where the constructive non-periodicity proof is lifted to the classical format

$$\neg \big(\exists T, P{>}0\ \forall t{\geq}T\ :\ c(t{+}P) = c(t)\big). \tag{12}$$

The explanation below remains constructively witnessable throughout.

> Question: "Does the center column always remain non-periodic?"
>
> Answer: Yes.
>
> Reason: The center column cannot become periodic or eventually periodic. From the center dynamics one defines a trace M_t. The result is tail anti-periodicity:
>
> $$\forall Q > 0,\ \exists t \geq T_Q \text{ such that } M_{t+Q} \neq M_t.$$

But eventual periodicity of the center column implies eventual periodicity of the marker trace, contradiction.

Remark 1.2.4. Proof theory is often associated with limitative results—this time, however, we believe it leads to a conclusive positive outcome.

This could be because the question has two advantages. First, it removes the combinatorial slack present in wider strip questions: the center column is a canonically distinguished observable, hence fixed by the standard single-seed initial condition and free from arbitrary window choices. Second, it brings the proof obligation into a form that is compatible with exact Δ_0 reasoning. Since Rule 30 is defined by a radius-1 update rule, any hypothetical eventual periodicity of the center column must be reflected in a bounded cone of overlapping local constraints around the center. This makes it possible to replace global spacetime speculation by finite witness analysis, i.e. accessible to logical methods.

Our Ansatz is therefore proof theoretic rather than statistical. We do not argue from apparent empirical simulations, nor from heuristic predecessor counts. Instead, we work within a computably enumerable (RE) theory [Turing, 1937; Kleene, 1952; Boolos et al., 2012] in a constructive [Troelstra and van Dalen, 1988] setting, reducing the question to a bounded witness structure consisting of a finite number of named overlap equations around the center cone.

The proof strategy is then a finite contradiction analysis. Each witness induces a small Boolean constraint system whose variables represent local predecessor bits and whose equations are exactly the relevant Rule 30 update clauses. The semantics of the problem are thus shifted from the infinite spacetime to a finite overlap calculus: if the center column were eventually periodic, then a corresponding bounded witness would exist; if every such witness is contradictory, then the periodicity hypothesis is impossible.

A central ingredient in making this reduction sound is finite-speed dependence [Hedlund, 1969; Wolfram, 1984]. After k time steps, the value of a cell depends only on the radius-k interval in the earlier row. Hence values outside the relevant backward cone cannot affect the local witness analysis. This locality barrier is what legitimizes bounded contradiction arguments: once a hypothetical center-periodic regime has been encoded into a finite witness, the witness may be analyzed in isolation, because unspecified data outside the cone are semantically irrelevant to the cells under consideration.

In the above sense, the center-periodicity problem serves both as a specific theorem target and as a test case for converting the informal complexity claims that have surrounded Rule 30 since its introduction [Wolfram, 1983, 2002] into exact finite constraint analyses that can be formalized and mechanically checked.

1.3 MECHANIZATION

Mechanization 1.3.1 (Proof environment)**.** The formal verification accompanying this paper is implemented in Rocq, whose trusted kernel is based on the Calculus of Inductive Constructions [Coquand and Huet, 1988]. The development is written in Gallina, the specification language of

and is publicly available as the project `T004` within the *repository* at

github.com/Milan-Rosko/Proofcase

The mechanization certifies the local and global infrastructure used in the proof of the main theorem. The development is therefore not a simulation of Rule 30 but a machine-checked proof that the logical reductions appearing in the manuscript are valid in the Rocq kernel.

Mechanization 1.3.2 (Formal layers)**.** The formal development is structured into logical layers.

Layer 1. Local Rule semantics.

This layer defines the local update rule

$$\mathcal{R}_{30}(a, b, c) = a \oplus (b \vee c) \tag{13}$$

and proves elementary properties of the rule used throughout the development. In particular, the complete truth-table characterization of Rule 30 is formalized and used as the basis for all local reasoning. The executable rule and its evaluation lemmas are defined in

`R00__Base.v.` (14)

Layer 2. Cone locality and evolution.

This layer formalizes finite-speed dependence of the cellular automaton. The key result states that after k time steps the value of a cell depends only on a radius-k interval in the earlier row. This property justifies restricting reasoning to bounded light cones around the center column; it is implemented in

`R02__Local_Lemmas.v.` (15)

Layer 3. *Canonical trajectory and derived traces.*

This layer constructs the canonical single-seed trajectory and defines derived sequences extracted from bounded neighborhoods of the center column. In particular, the development defines a compressed trace (the *marker trace*) obtained from the center dynamics. The trace is designed so that periodic behavior of the center column would imply periodicity of the marker trace. These constructions appear in

`R07__Packed_Compression.v.` (16)

Layer 4. Anti-periodicity and strip reduction.

The final layer proves that the marker trace is tail anti-periodic:

$$\forall P > 0, \quad \exists t \quad M_{t+P} \neq M_t. \tag{17}$$

Since the marker trace is determined by the center dynamics, this immediately implies that the center column itself cannot be eventually periodic. The result is then lifted to all centered strips

by a simple induction argument: eventual periodicity of a larger strip would imply eventual periodicity of the center column. It is proven in

`R09__Compressed_Not_Eventually_Periodic.v` (18)

and the strip-level theorem is assembled in

`R06__No_Strip_Periodicity.v.` (19)

Mechanization 1.3.3 (Artifact)**.** The repository is designed as a checkable artifact and is also available in OCaml. A complete build of the project compiles the Rocq theories and verifies all theorems using the Rocq kernel:

(i) the source files formalizing $\mathcal{R}_{30}$ dynamics and the proof infrastructure;

(ii) the build scripts required to compile the development;

(iii) the kernel verification log confirming that all theorems are accepted by a type checker.

The artifact should therefore be interpreted as a certified mathematical argument rather than as an external computational context.

The mechanized development certifies the exact mathematical endpoint claimed in this paper: that the canonical single-seed Rule 30 trajectory contains no eventually periodic centered strip of finite width. Informal discussion in the manuscript concerning broader implications for the global dynamics should be interpreted as heuristic motivation unless a corresponding theorem appears explicitly in the formal development.

The endpoint of our development is the FOL theorem

$$\forall m \geq 0, \quad \neg\exists T, P > 0 \quad \forall t \geq T, \quad B_{t+P}^{(m)} = B_t^{(m)}. \tag{20}$$

In words: no centered finite strip in the canonical Rule 30 trajectory is eventually periodic. This theorem is assembled in the file

`R06__No_Strip_Periodicity.v` (21)

from the anti-periodicity theorem for the marker trace together with elementary strip-reduction lemmas.

Mechanization 1.3.4 (Trust)**.** The trusted base of the result is the standard one for a Rocq-verified development: the Rocq kernel, the parser and build system, and the surrounding compilation environment.

The constructive core of the development uses no axioms beyond those of the Calculus of Inductive Constructions. The classical rendering (Corollary 3.1.2) additionally imports the law of the excluded middle via `Classical_Prop.classic`; this is the sole extra-logical dependency.

Consequently, the repository provides a complete kernel-checked proof of the main theorem presented in this paper.

2 MAIN THEOREM

2.1 PROGENITOR THEORY

Mechanization 2.1.1 (`R00,R01,R02`)**.** The theorem chain in this subsection is formalized in:

```
R00__Base.v
R01__Seed.v
R02__Local_Lemmas.v
```
(22)

Definition 2.1.2 (Orientation)**.** Our first intuition is that values from different rows may be related only through explicit $\mathcal{R}_{30}$ equations. Let

$$u_{t,x} \in \{0,1\}. \tag{23}$$

In the mechanization, the seed row $t = 0$ is placed at position 5:

$$\cdots \quad 0 \quad 0 \quad 0 \quad 0 \quad \underbrace{1}_{x=5} \quad 0 \quad 0 \quad 0 \quad 0 \quad \cdots \tag{24}$$

This is a mechanization convention; translation invariance (Origo, below) lets us shift the seed to the origin without loss of generality. In the local window $x = 3, 4, 5, 6, 7$,

$$u_{0,3} = 0, \quad u_{0,4} = 0, \quad u_{0,5} = 1, \quad u_{0,6} = 0, \quad u_{0,7} = 0. \tag{25}$$

Definition 2.1.3 (Progenitor)**.** A configuration

$$u = (u_x)_{x\in\mathbb{Z}} \in \{0,1\}^{\mathbb{Z}} \tag{26}$$

is a *progenitor* of a configuration $v = (v_x)_{x\in\mathbb{Z}} \in \{0,1\}^{\mathbb{Z}}$ under $\mathcal{R}_{30}$ if

(i) u is finitely supported, and

(ii) for every $x \in \mathbb{Z}$,

$$v_x = \mathcal{R}_{30}(u_{x-1}, u_x, u_{x+1}). \tag{27}$$

2.2 ORIGINAL SIN THEOREM

Remark 2.2.1. Our first theorem considers the Garden of Eden concept in cellular automata [Moore, 1962; Myhill, 1963]: a configuration that admits no predecessor under the global map. Here the single-seed row is orphan-like, and the obstruction is inherited by every subsequent generation.

Mechanization 2.2.2 (`R01,R02,R03`)**.** The theorem chain in this subsection is formalized in:

```
R01__Seed.v
R02__Local_Lemmas.v
R03__Periodicity.v
```
(28)

Lemma 2.2.3 (Truth table of $\mathcal{R}_{30}$)**.** Consider the table of $a \oplus (b \vee c)$,

a	b	c	$b \vee c$	$\mathcal{R}_{30}(a,b,c)$
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	1	1
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	0

The table partitions all triples into those producing 0 and those producing 1. All later predecessor arguments depend only on these eight cases. By logic,

$$\mathcal{R}_{30}^{-1}(0) = \{000, 101, 110, 111\}, \quad \mathcal{R}_{30}^{-1}(1) = \{001, 010, 011, 100\}. \tag{29}$$

Lemma 2.2.4 (Origo)**.** By translation invariance of $\mathcal{R}_{30}$, we may shift coordinates so that the unique seed lies at the origin. Thus, without loss of generality,

$$u_{0,x} = \begin{cases} 1 & x = 0, \\ 0 & x \neq 0. \end{cases} \tag{30}$$

Theorem 2.2.5 (Original Sin)**.** The single-seed row has no valid progenitor under $\mathcal{R}_{30}$.

Proof. We proceed by *reductio ad absurdum.* Assume a progenitor exists at row -1,

$$p = u_{-1,-2}, \tag{31}$$
$$q = u_{-1,-1}, \tag{32}$$
$$r = u_{-1,0}, \tag{33}$$
$$s = u_{-1,1}, \tag{34}$$
$$e = u_{-1,2}, \tag{35}$$
$$v = u_{-1,3}, \tag{36}$$
$$w = u_{-1,4}. \tag{37}$$

The seed row implies the five Rule 30 constraints

$$\mathcal{R}_{30}(p,q,r) = 0, \tag{38}$$
$$\mathcal{R}_{30}(q,r,s) = 1, \tag{39}$$
$$\mathcal{R}_{30}(r,s,e) = 0, \tag{40}$$
$$\mathcal{R}_{30}(s,e,v) = 0, \tag{41}$$
$$\mathcal{R}_{30}(e,v,w) = 0. \tag{42}$$

Since the progenitor is finitely supported, sufficiently far from the seed all predecessor cells are 0. Local propagation therefore forces the outer predecessor bits. From the left tail:

$$\mathcal{R}_{30}(0,0,p) = 0 \implies p = 0, \tag{43}$$
$$\mathcal{R}_{30}(0,p,q) = 0 \implies q = 0. \tag{44}$$

From the right tail:

$$\mathcal{R}_{30}(w, 0, 0) = 0 \implies w = 0, \tag{45}$$
$$\mathcal{R}_{30}(v, w, 0) = 0 \implies v = 0. \tag{46}$$

Substituting these values yields

$$\mathcal{R}_{30}(0, 0, r) = 0, \tag{47}$$
$$\mathcal{R}_{30}(0, r, s) = 1, \tag{48}$$
$$\mathcal{R}_{30}(r, s, e) = 0, \tag{49}$$
$$\mathcal{R}_{30}(s, e, 0) = 0, \tag{50}$$
$$\mathcal{R}_{30}(e, 0, 0) = 0. \tag{51}$$

From the first and last equations we obtain

$$r = 0 \wedge e = 0 \implies \mathcal{R}_{30}(0, 0, s) = 1 \implies s = 1. \tag{52}$$

Which is a contradiction,

$$\mathcal{R}_{30}(r, s, e) = \mathcal{R}_{30}(0, 1, 0) = 1 \neq 0 \implies \perp \tag{53}$$

Therefore no progenitor row exists. ■

2.3 COMPRESSED GLITCH THEOREM

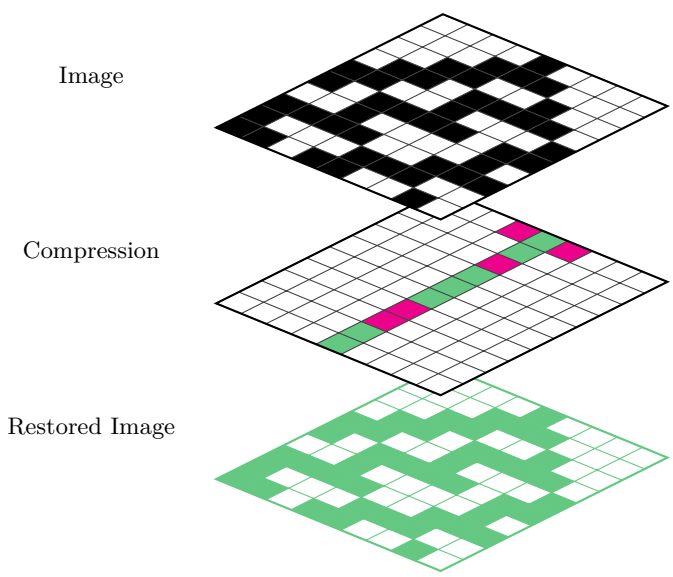

Figure 3. Reconstruction of the $\mathcal{R}_{30}$ spacetime from a T-shaped compressed seed: the initial triple $(0, 1, 0)$ and the center column $c(t)$. Because $\mathcal{R}_{30}$ is deterministic, the full spacetime can be propagated outward from this minimal vertical–horizontal skeleton in any direction, so the center column together with the seed row determines the entire trajectory.

Remark 2.3.1. By Theorem 2.2.5, the local contradiction is unconditional. We now pass from the center column to arbitrary centered finite strips.

Definition 2.3.2 (Centered strip, projection, and eventual periodicity)**.** For $m \geq 0$ define

$$B_t^{(m)} = (u_{t,-m}, u_{t,-m+1}, \ldots, u_{t,m}). \tag{54}$$

For $m \geq 0$, define the projection map

$$\pi_m\big(B_t^{(m+1)}\big) := B_t^{(m)}, \tag{55}$$

i.e. π_m deletes the two outer coordinates. We call $B_t^{(m)}$ eventually periodic if there exist integers $T \geq 0$ and $P > 0$ such that

$$B_{t+P}^{(m)} = B_t^{(m)} \quad \text{for all } t \geq T. \tag{56}$$

Lemma 2.3.3 (Projection of an eventually periodic strip)**.** If $B_t^{(m+1)}$ is eventually periodic, then $B_t^{(m)}$ is eventually periodic.

Proof. If there exist $T \geq 0$ and $P > 0$ such that

$$B_{t+P}^{(m+1)} = B_t^{(m+1)} \quad \text{for all } t \geq T, \tag{57}$$

then applying π_m on both sides yields

$$B_{t+P}^{(m)} = B_t^{(m)} \quad \text{for all } t \geq T. \tag{58}$$

■

Remark 2.3.4. We call this obstruction a *compressed glitch*: any eventual periodicity on a larger centered strip must survive repeated projection to smaller strips, but the projection chain terminates in a contradiction at the center.

Mechanization 2.3.5 (`R07,R09,R06`)**.** The theorem chain in this subsection is formalized in:

```
R07__Packed_Compression.v
R09__Compressed_Not_Eventually_Periodic.v
R06__No_Strip_Periodicity.v
```
(59)

Theorem 2.3.6 (Compressed Glitch)**.** For every $m \geq 0$, the centered strip $B_t^{(m)}$ is not eventually periodic.

$$\forall m \geq 0, \quad \neg\exists T, P > 0 \quad \forall t \geq T, \quad B_{t+P}^{(m)} = B_t^{(m)}. \tag{60}$$

Proof sketch. By *reductio ad absurdum.* Assume toward contradiction that

$$S := \{m \in \mathbb{N} \mid B^{(m)} \text{ is eventually periodic}\} \tag{61}$$

is nonempty, and let $m_* := \min S$, so that

$$m_* = 0 \implies B_t^{(0)} = (u_{t,0}) \tag{62}$$

is eventually periodic. But the bridge claim

$$\text{center periodic} \implies \text{marker periodic} \tag{63}$$

as in Mechanization 2.3.5. The marker trace is tail anti-periodic, again by the references listed there. This contradiction shows $m_* \neq 0$, hence $m_* \geq 1$. Since $m_* \in S$, the strip $B^{(m_*)}$ is *eventually periodic*. By Lemma 2.3.3, $B^{(m_*-1)}$ is *eventually periodic*, so $m_* - 1 \in S$, contradicting minimality of m_*. Hence $S = \varnothing$, and every centered strip is not eventually periodic. ■

Remark 2.3.7. Consequently, *mixed periodicity* cannot occur: any eventually periodic regime on a larger centered strip projects to eventual periodicity on every smaller centered strip. The strip-level endpoint is the third entry in Mechanization 2.3.5. In particular, setting $m = 0$ yields non-eventual periodicity of $c(t) = u_{t,0}$, the non-periodicity result used in Lemma 3.2.2 and Corollaries 3.2.3–3.2.4.

3 COROLLARY

3.1 TRANSLATION

Thesis 3.1.1 (Constructive-to-classical bridge)**.** The obstruction argument is finitary throughout: bounded witness extraction, local contradiction, and strip projection are carried out inside CIC. The manuscript statement is then presented in standard classical syntax by one discharge step using `Classical_Prop.classic`.

Corollary 3.1.2 (Classical aperiodicity)**.** By Theorem 2.3.6, in particular the instance $m = 0$, the center column admits no eventually periodic tail. In classical form:

$$\neg\big(\exists T, P{>}0 \;\; \forall t{\geq}T \;:\; c(t{+}P) = c(t)\big). \tag{64}$$

Since intuitionistic logic is a subsystem of classical logic, the constructive proof remains valid classically. The only additional dependency is

$$\texttt{Classical_Prop.classic}, \tag{65}$$

which supplies LEM at the unique point where the constructive proof is lifted to the classical existential format.

3.2 COMPLEXITY

Definition 3.2.1 (Crossing complexity)**.** Fix a deterministic, advice-free local sequential model $\mathcal{M}$ that computes the center bit $c(n) = u_{n,0}$. For each input depth n and each cut through the backward light cone of the center computation, a *crossing state* is the finite summary of all information that passes from the processed side of the cut to the unprocessed side. Write

$$\mathrm{Cross}_{\mathcal{M}}(n) \tag{66}$$

for the set of all crossing states reachable in exact runs of $\mathcal{M}$ on inputs of depth n, and define the *crossing complexity* by

$$\chi_{\mathcal{M}}(n) := \big|\mathrm{Cross}_{\mathcal{M}}(n)\big|. \tag{67}$$

Lemma 3.2.2 (Crossing complexity)**.** Let $\mathcal{M}$ be a deterministic, advice-free local sequential algorithm that exactly computes $c(n)$. If the crossing complexity is uniformly bounded,

$$\exists K \ \ \forall n, \quad \chi_{\mathcal{M}}(n) \leq K, \tag{68}$$

then the center column is eventually periodic.

Proof sketch. Fix a bound K. Since $\mathcal{M}$ is deterministic and advice-free, the future behavior beyond any cut is determined by the current crossing state. Along increasing depths, $\mathcal{M}$ induces a deterministic evolution on a finite set of at most K states. A deterministic evolution on a finite state space must eventually enter a cycle: there exist integers $T \geq 0$ and $P > 0$ such that the crossing state at depth $n + P$ equals the crossing state at depth n for all $n \geq T$. Because the output bit is a deterministic function of the terminal state,

$$c(n + P) = c(n) \quad \text{for all } n \geq T. \tag{69}$$

■

Corollary 3.2.3 (Unbounded crossing complexity)**.** Combining Lemma 3.2.2 with Theorem 2.3.6, for every exact deterministic advice-free local sequential algorithm $\mathcal{M}$ for $c(n)$,

$$\forall K \ \ \exists n, \quad \chi_{\mathcal{M}}(n) > K. \tag{70}$$

The crossing complexity must diverge.

Corollary 3.2.4 (Linear time lower bound)**.** For any concrete machine model in which sublinear time implies bounded crossing complexity,

$$\mathrm{Time}_{\mathcal{M}}(n) = \Omega(n). \tag{71}$$

The chain of implications follows from $\mathrm{Time}(n) = o(n)$ and Theorem 2.3.6,

$$o(n) \implies \chi_{\mathcal{M}}(n) = O(1) \implies \text{eventually periodic center} \implies \bot. \tag{72}$$

Remark 3.2.5. The crossing-complexity obstruction established above bears on the Twenty Problems of Wolfram [1985, Problem 5, 8, 15, 16, 18, 20] and [Wolfram, 2002, Chapter 7], which ask how $\mathcal{R}_{30}$ relates to randomness—understood here as the inability of any finite description to compress or predict the output sequence. Our CIC kernel and the constraint calculus show that the center column cannot eventually repeat. Because eventual periodicity is the simplest form of finite-state compressibility, its refutation must supply a necessary condition for any semantics of randomness: a sequence that eventually repeats is certainly not random. In this sense, the first-order obstruction proved here meets one requirement of HOL, namely that the center column resists the most elementary form of algorithmic compression.

4 INTERPRETATION

4.1 METHOD

Remark 4.1.1. The methodology is organized around bounded witness structures: any hypothetical eventual periodicity of the center column induces a finite system of overlapping $\mathcal{R}_{30}$ equations around the center cone, and the proof proceeds by exhaustive contradiction analysis within this system. Every refutation step uses only named $\mathcal{R}_{30}$ overlap equations, truth-table exclusions, and bounded locality. This provides a proof-theoretic architecture for non-periodicity results, replacing global spacetime speculation with finite constraint analysis of the kind surveyed in Wolfram [1985] and Wolfram [2002, Chapter 7].

4.2 REALIZABILITY

Remark 4.2.1. The proof strategy has its roots in the realizability tradition of Kleene [1952] and the Brouwer–Heyting–Kolmogorov (BHK) interpretation [Troelstra and van Dalen, 1988]. Under BHK, a proof of $\neg\varphi$ is a construction that transforms any hypothetical realizer of φ into a contradiction. This is precisely what the bounded witness method delivers: given any candidate realizer of eventual periodicity—a pair (T, P) together with the induced overlap system—the construction produces an explicit $\perp$ from the local $\mathcal{R}_{30}$ constraints. The refutation is therefore not a classical existence argument but a constructive procedure that maps witnesses to contradictions, exactly as BHK semantics requires. More precisely: assume the Σ^0_2 hypothesis $(\exists T, P\, \forall t{\geq}T \ldots)$, extract from the existential witnesses a Π^0_1 constraint system, and show it unsatisfiable by Δ_0 case analysis.

4.3 THE XOR

Remark 4.3.1. The connective that makes this refutation rigid is $\oplus$ (XOR). Among the sixteen binary Boolean functions, $\oplus$ is the archetypical metatheoretic connective: it is the only non-constant, non-monotone, self-inverse Boolean operation, and it is the gate through which parity enters the constraint system. In the $\mathcal{R}_{30}$ update rule $a \oplus (b \vee c)$, the disjunctive component $b \vee c$ propagates information monotonically, but the outer $\oplus$ inverts the parity at each step. A periodic regime would require the local overlap equations to close consistently across the period boundary; $\oplus$ prevents this by flipping the parity balance whenever the monotone core stabilizes. It is this parity obstruction that makes every candidate witness contradictory and distinguishes $\mathcal{R}_{30}$ from rules whose local maps are monotone or affine.

The connection to Kolmogorov complexity [Kolmogorov, 1965; Li and Vitányi, 2008] is immediate: an eventually periodic sequence admits a finite generating program, so refuting eventual periodicity establishes that the center column is not compressible by periodic generators. We do not claim this settles the full Kolmogorov Complexity of the center column—that would require excluding all finite programs—but it provides a first certified lower bound, proved by finitary methods traceable to Gödel [1931] and Kleene [1952].

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Milan Rosko is from
University of Hagen, Germany

ORCID: 0009-0003-1363-7158
Email: Q1012878@studium.fernuni-hagen.de
Email: hi-at-milanrosko.com

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